

SRSI RESEARCH PROJECT

Occlusion-Adaptive Temporal Memory

Giving self-driving cars short-term memory for hidden objects

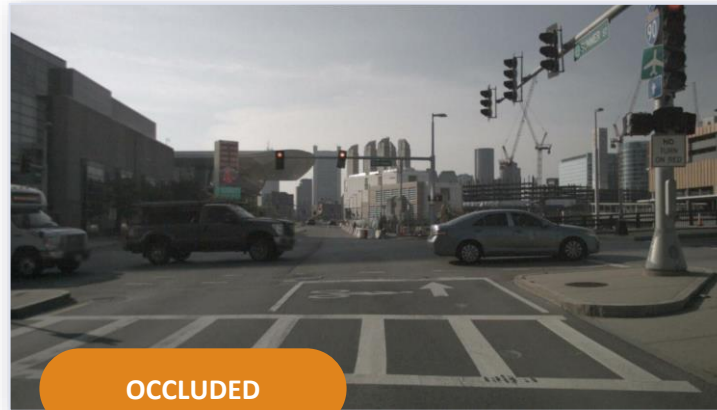
nuScenes CAM_FRONT — an autonomous-vehicle perception research project

THE PROBLEM IN AUTONOMOUS DRIVING

Single-frame detectors forget what they can't see



Car clearly visible



Fully hidden from view



Reappears down the road

A detector that only looks at the middle frame has no idea a car is even there.

"Like watching someone walk behind a parked bus — you don't assume they vanished."

THE DATASET

nuScenes — Front Camera Only

1

nuScenes: a large-scale AV dataset with synchronized cameras, LiDAR, radar, and 3D box annotations that include a per-object visibility rating

2

This project uses only CAM_FRONT — one monocular camera, no LiDAR or other views feeding detection itself

3

Front-only keeps the pipeline laptop-scale and isolates the occlusion-memory problem, instead of mixing in multi-camera fusion

4

LiDAR and the 3D annotations aren't wasted, though — they come back as independent ground truth for evaluating OATM (next: Evaluation Methodology)



CAM_FRONT — the only view this project detects from

THE CORE RESEARCH QUESTION

Can short-term visual memory help a camera keep track of temporarily hidden road users — without inventing false ones?

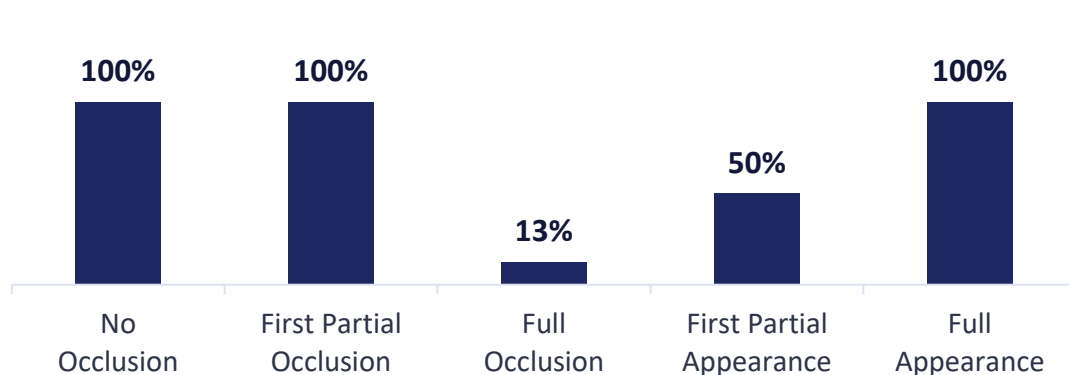
Hypothesis

Occlusion-aware temporal memory preserves more valid tracks through short occlusions than single-frame detection or conventional tracking — without creating an unacceptable number of false, ghost tracks.

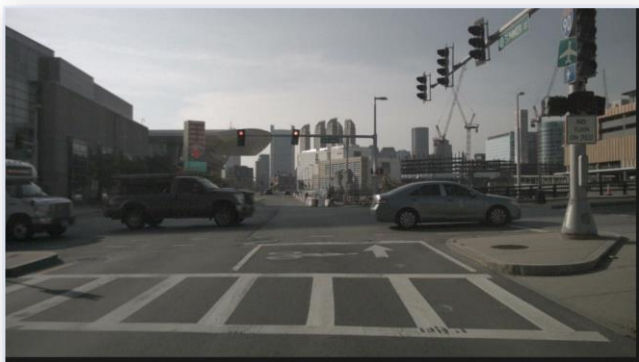
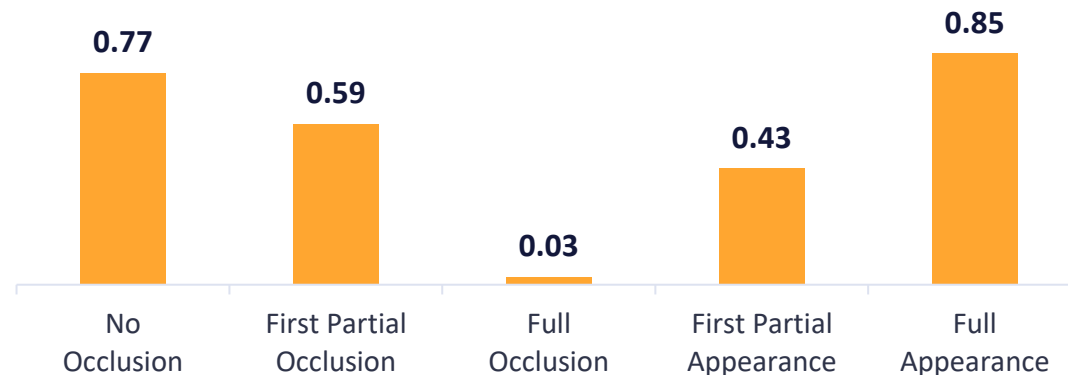
Detection Collapses at Full Occlusion

Pretrained YOLO nano, CPU-only, on 8 real nuScenes occlusion sequences — tracked through 5 stages, from clearly visible to fully hidden and back.

Detection rate by stage



Mean confidence by stage (0 = undetected)



Full occlusion, sample_003 — YOLO draws zero boxes on the target. Not misdetected — simply absent.

100% → 12%

detection rate at full occlusion

0.77 → 0.03

mean confidence at full occlusion

100%

correct class whenever detected

Partial-occlusion stages have small sample sizes (n=2-3) — interpret with caution.

The Simplest Fix: Freeze the Last Box

When YOLO loses the target, keep drawing its last known box instead of letting it vanish.

BEST CASE



sample_003 — reappears next to a bus (83 px, IoU 0.34 — boxes barely overlap by a third)

Also tested: sample_001 (127px), sample_004 (423px), sample_005 (432px) — same pattern across all 5.

WORST CASE



sample_006 — crossing an intersection (589 px, IoU 0.00)

331 px

mean center error on reappearance

0.085

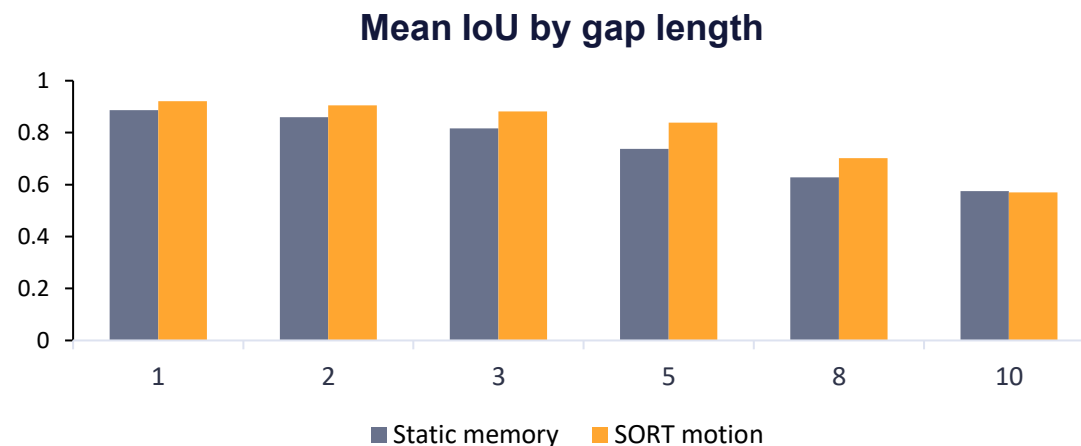
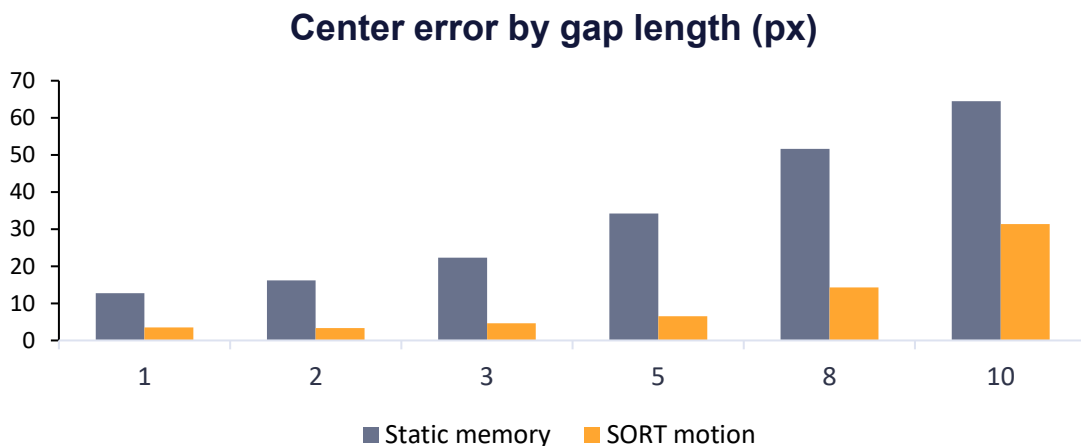
mean IoU (median: 0.000)

3 / 5

samples had zero box overlap

Motion Prediction Helps — With Caveats

A mini SORT-style tracker predicting velocity instead of freezing position — tested on 9 real car tracks.



46 → 17 px

mean center error, static → SORT

0.673 → 0.720

mean IoU, static → SORT

9 tracks

cars only, across 3 clips

THE CATCH

Almost the entire win comes from 2 of 9 tracks — on the dominant clip, static memory ties SORT. The reference boxes were Kalman-corrected (not raw YOLO), and long gaps bypass the tracker's own expiry rule. A real demonstration of motion extrapolation — not yet an independent validation of SORT.

THE DIAGNOSIS

A static memory box goes stale because it ignores motion.

Real objects keep moving while hidden. A memory system that only remembers where something was will drift further wrong the longer it stays hidden.

THE GAP IN EXISTING TOOLS

Single-frame detection

no memory at all — occluded means gone

SORT

cuts error 46→17px on smooth tracks — but N=9 cars, circular reference

ByteTrack

still needs some visible evidence to link to

Fixed temporal memory

blind to motion and its own ghost-track risk

None decide hidden vs. lost vs. gone, or decay confidence accordingly — that's the gap OATM is built to fill.

THE PROPOSED FIX

OATM: Occlusion-Adaptive Temporal Memory

1

Appearance memory: stores the last clear crop of the object, so it can be re-identified on reappearance

2

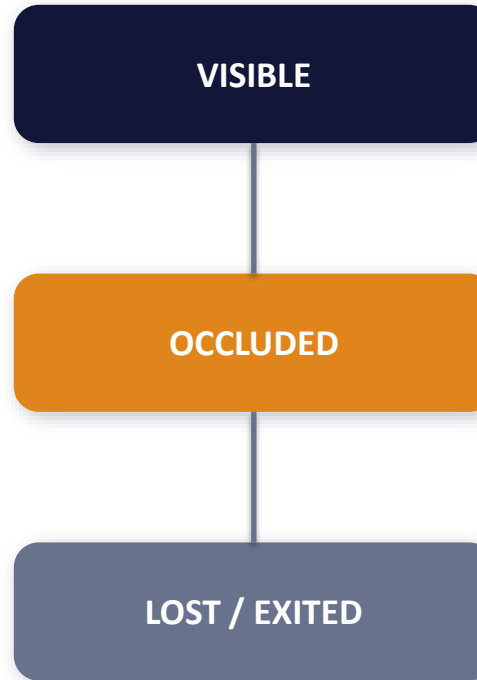
Motion memory: a Kalman-style predict → observe → correct loop — the same mechanism already prototyped in Evidence 3, fixed to use an independent reference and real track-expiry

3

Occlusion classifier: VISIBLE / OCCLUDED / LOST / EXITED — decides how much to trust the motion prediction right now

4

Confidence decays smoothly, not on a fixed countdown; anti-ghost termination retires a track before it becomes a phantom



reappearance → back to VISIBLE

Ground Truth From LiDAR, Not the Camera Itself

1

Use nuScenes' 3D box annotations + LiDAR point cloud as ground truth, projected into CAM_FRONT — never derived from the front camera's own detections

2

Run OATM exactly as designed: CAM_FRONT only, genuinely blind to the object during its occluded frames

3

Score OATM's predicted position during occlusion against that projected LiDAR / 3D ground truth, not against another camera-based tracker

4

This directly fixes the circularity in Evidence 3, where the reference boxes came from the same Kalman model being tested

The front camera can't see through the occlusion — but LiDAR still can. That's what makes this an independent check.

HOW OATM COMPARES

A Design Comparison

OATM has not been implemented or measured yet — this is a design-level comparison.

Method	How it works	Main weakness
Single-frame detection	Looks only at the current image	Hidden object instantly disappears
SORT	Predicts motion — measured 46→17px error on 9 tracks	Tiny sample; reference partly circular; expiry bypassed
ByteTrack	Links strong and weak detections	Still needs some visible evidence
Fixed temporal memory	Keeps missing objects for N frames	May forget early or keep ghosts too long
OATM (proposed)	Appearance + motion + occlusion-aware decay	More complex; needs careful tuning

ROADMAP

Proving It

1

Build natural + controlled occlusion test sets from nuScenes

2

Fix SORT's circular reference and disabled track-expiry, add ByteTrack, then benchmark all 5 against OATM

3

Metrics: occluded-object recall, identity preservation, ghost-track rate, recovery time

4

Split train / validation / test by scene, not by frame, to avoid leakage

5

Expand from vehicle-only tracking to pedestrians and other vulnerable road users

Thank you — questions welcome

REFERENCES

Papers We Read

Ma, J., Yang, J., Fu, F., Hao, D., & Wang, R. (2026). *Occlusion-aware multi-camera 3D multi-object tracking with dual-appearance dynamic association*. Applied Soft Computing.

Pang, Z., Li, J., Tokmakov, P., Chen, D., Zagoruyko, S., & Wang, Y.-X. (2023). *Standing Between Past and Future: Spatio-Temporal Modeling for Multi-Camera 3D Multi-Object Tracking*. CVPR 2023. arXiv:2302.03802.

Anwar, N., Bilodeau, G.-A., & Bouachir, W. (2024). *STF: Spatio-Temporal Fusion Module for Improving Video Object Detection*. arXiv:2402.10752.

Cai, J., Xu, M., Li, W., Xiong, Y., Xia, W., Tu, Z., & Soatto, S. (2022). *MeMOT: Multi-Object Tracking with Memory*. CVPR 2022, pp. 8080-8090.

Tokmakov, P., Li, J., Burgard, W., & Gaidon, A. (2021). *Learning to Track with Object Permanence*. ICCV 2021.

Geiger, A., Lenz, P., & Urtasun, R. (2012). *Are We Ready for Autonomous Driving? The KITTI Vision Benchmark Suite*. CVPR 2012, pp. 3354-3361.

Iftikhar, S., Zhang, Z., Asim, M., Muthanna, A., Koucheryavy, A., & Abd El-Latif, A. A. (2022). *Deep Learning-Based Pedestrian Detection in Autonomous Vehicles: Substantial Issues and Challenges*. Electronics, 11(21), 3551.